

REPLY TO PROFESSOR COSTALL

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ABSTRACT: A reply to the review of *Radical Embodied Cognitive Science* by Alan Costall.
Key words: Costall, *Radical Embodied Cognitive Science*, Turvey, Shaw

Because Professor Costall's review focuses largely on what is not in my book, I will briefly describe what the book is about, and why I wrote it. The purpose of the book is to introduce philosophers (of mind, of science, of cognitive science) to an ongoing research program that combines nonlinear dynamical modeling and Gibson's ecological psychology. Most of these philosophers view psychology very narrowly and are not familiar with this research program, despite its robustness. They also mostly believe that Noam Chomsky killed behaviorism dead in 1957. Like the ongoing research by behaviorists, the research program that I call "radical embodied cognitive science" has important implications for philosophical questions. These implications stem primarily from the fact that radical embodied cognitive science begins by rejecting a core principle of the Western tradition, namely that our knowledge and experience of the world is mediated by mental representations. Most philosophers take representationalism to be an *a priori* truth. So the purpose of the book was primarily to say that it is not only possible to do psychology without invoking mental representations, but also that there is a thriving research program that does just that, and is doing it with great success.

With that out of the way, I can address some of Professor Costall's specific complaints. Professor Costall wonders why my book is about cognition, with all of that term's intellectualist baggage. The easy answer is that there are departments and programs in what has come to be called "cognitive science" at universities all over the world. There is the Cognitive Science Society with an annual meeting at which self-identified cognitive scientists gather to share data. Cognitive science, as the kids say, is *a thing*. What unites cognitive scientists is not a commitment to intellectualism, but rather a commitment to computational and mathematical modeling. In 2013, the Cognitive Science Society awarded its most prestigious award, the Rummelhart Prize, to Linda Smith, who identifies herself as a cognitive scientist, but regularly makes a point of insisting that cognitive scientists don't have to be cognitivists. Many self-identified cognitive scientists are not intellectualists and do not believe that thinking is the computational manipulations of representations. In fact, many philosophers have accused me, and other non-

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cognitivist cognitive scientists, of changing the meaning of the term “cognition” so that it means intelligent or adaptive behavior. I do so explicitly in my book. The reason that it is worth keeping the term is that it allows us to keep the explanatory target and to say “See all that stuff that is called ‘cognition’ and supposedly requires complicated internal computational procedures? Well, we can explain that stuff without the complicated internal computational procedures.”

Professor Costall also complains that I focus too much on what he calls “GATTS,” for “Gibson according to Turvey and Shaw,” and ignore the work of sociologists and other Gibsonians. Professor Costall is right that I focus on a particular variety of post-Gibsonian ecological psychology, one which originated at the University of Connecticut and Haskins Labs and for which Michael Turvey and Robert Shaw are among the founding figures. Why would I choose to focus on this variety of post-Gibsonian ecological psychology? The most obvious reason would be that it is this variety that has been the most successful as a science. The ecological psychologists at the University of Connecticut have received millions of dollars in research funding, placed scores of articles in top journals, and trained dozens of Ph.D. and post-doctoral students who have gone on to found their own labs to practice what Professor Costall calls GATTS. Moreover, these practitioners of GATTS, by and large, consider themselves to be non-cognitivist cognitive scientists, and they regularly publish their work in journals (such as *Cognition*, *Cognitive Science*, and *Topics in Cognitive Science*) aimed at other cognitive scientists, cognitivist and otherwise.

I hope readers will agree that “radical embodied cognitive science” is a better, more mellifluous name than “GATTS,” and moreover that there is more to radical embodied cognitive science than Gibson, Turvey, and Shaw. There is also a lot of Scott Kelso and Brian Cantwell Smith and Ian Hacking and David Lee and. . . In any event, it seems to me that introducing a successful research program in cognitive science to philosophers, showing how it really explains activity that cognitive scientists are interested in explaining without invoking internal representations, and setting out some of the philosophical consequences of the research program is plenty of work for a book to do. It is for this reason that there was no discussion of the sociologists of situatedness of the 1980s and 1990s that Professor Costall mentions. Sociology, even sociology done by fellow travellers, didn’t shape my thinking on these issues and seemed too far afield for discussion. Professor Costall also complains that I neglect the early participants in post-Gibsonian ecological psychology, those that contributed only before the International Society for Ecological Psychology solidified around GATTS. As with the sociologists, I had read some of the work of these figures while in graduate school, but they have had very little influence on my subsequent thinking. Indeed, these early contributors have had very little recent influence on ecological psychology or on what I call radical embodied cognitive science. Readers of Thomas Kuhn will know why this is the case.